# Laboratory ECC (second part)

## Elliptic Curve: Projective Coordinates

***Task 1: Based on the class ProjectivePCurve define the algebraic sum and the scalar multiplication of points over an elliptic curve.***

- ***These operations must be defined using __add__ and __rmul__ methods. The scalar multiplication must be defined efficiently!***
- ***In order to define the mentioned operations, you can use Sagemath aiming to get the corresponding elliptic curve but you can not use the Sagemath operations over the elliptic curve.***
- ***You can extend the class as you pleased, with any attribute, method or auxiliary function.***
- ***Modular arithmetic must be achieved via the basic Python operations.***
- ***Remember that different projectives coordinates could represent the same point over the curve.***

Elliptic curve operation defined by Sagemath may be useful so as to test and compare your implementation. Ones your implementation is done, base on the elliptic curve:

$$C = EllipticCurve(Zmod(23), [0x1, 0x1])$$

An the points:

$$P = C(10,8,2)$$
$$Q = C(9,30,3)$$
$$R = C(9,39,3)$$

Test:

$$303 * P$$
$$Q + R$$
$$P + Q$$

Compare your implementation results against the Sagemath implementation, they must be the same, it may be useful to compare points based on its affine representation.